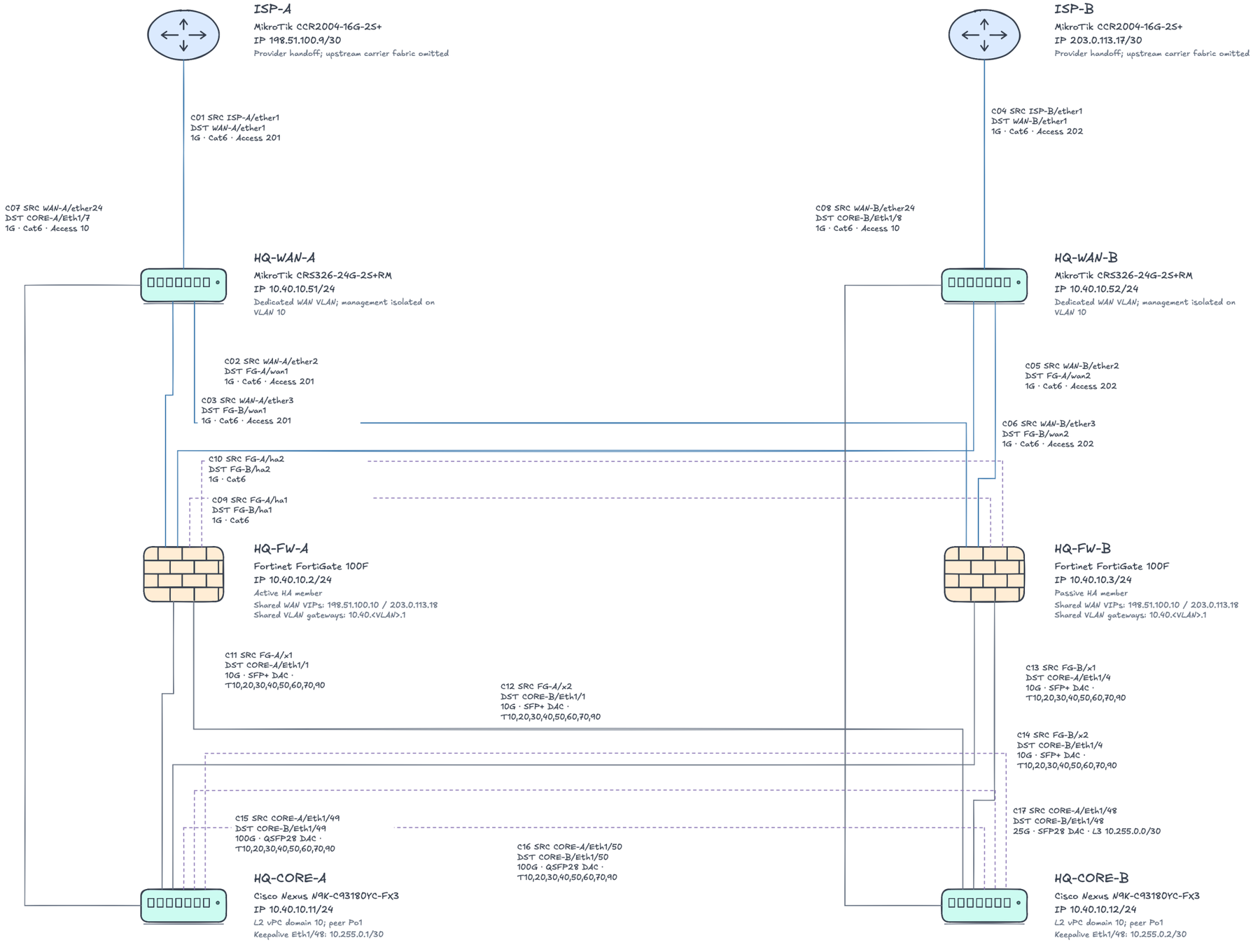
AURORA / HQ perimeter & core

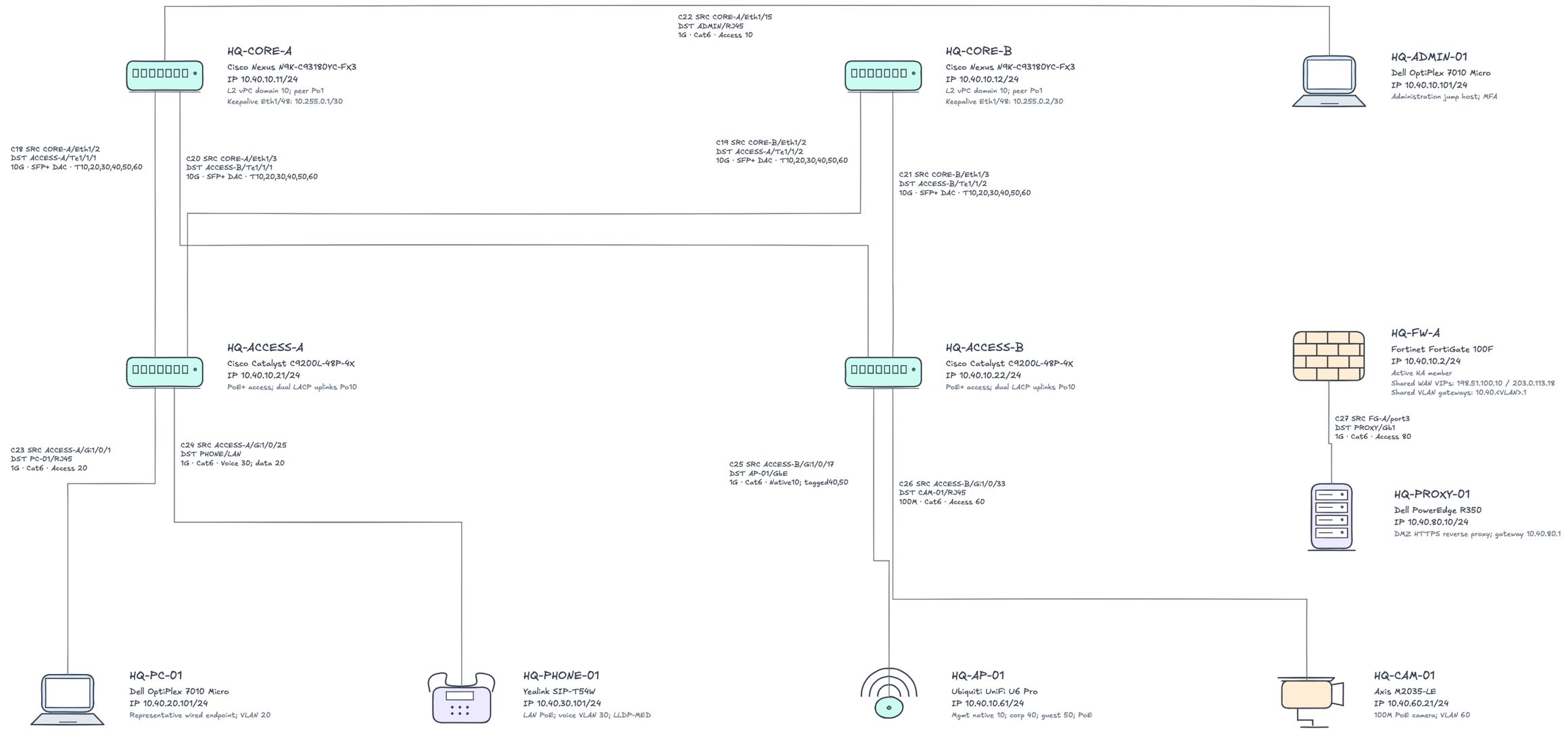
Real example hardware · fictional addressing · WAN VIPs on both HA members · vPC core · local cables only



WAN-A VLAN201 and WAN-B VLAN202 carry public handoffs only; VLAN10 administration is isolated. HA uses ha1 + ha2; core Po1 uses two 100G links.

AURORA / HQ access & DMZ

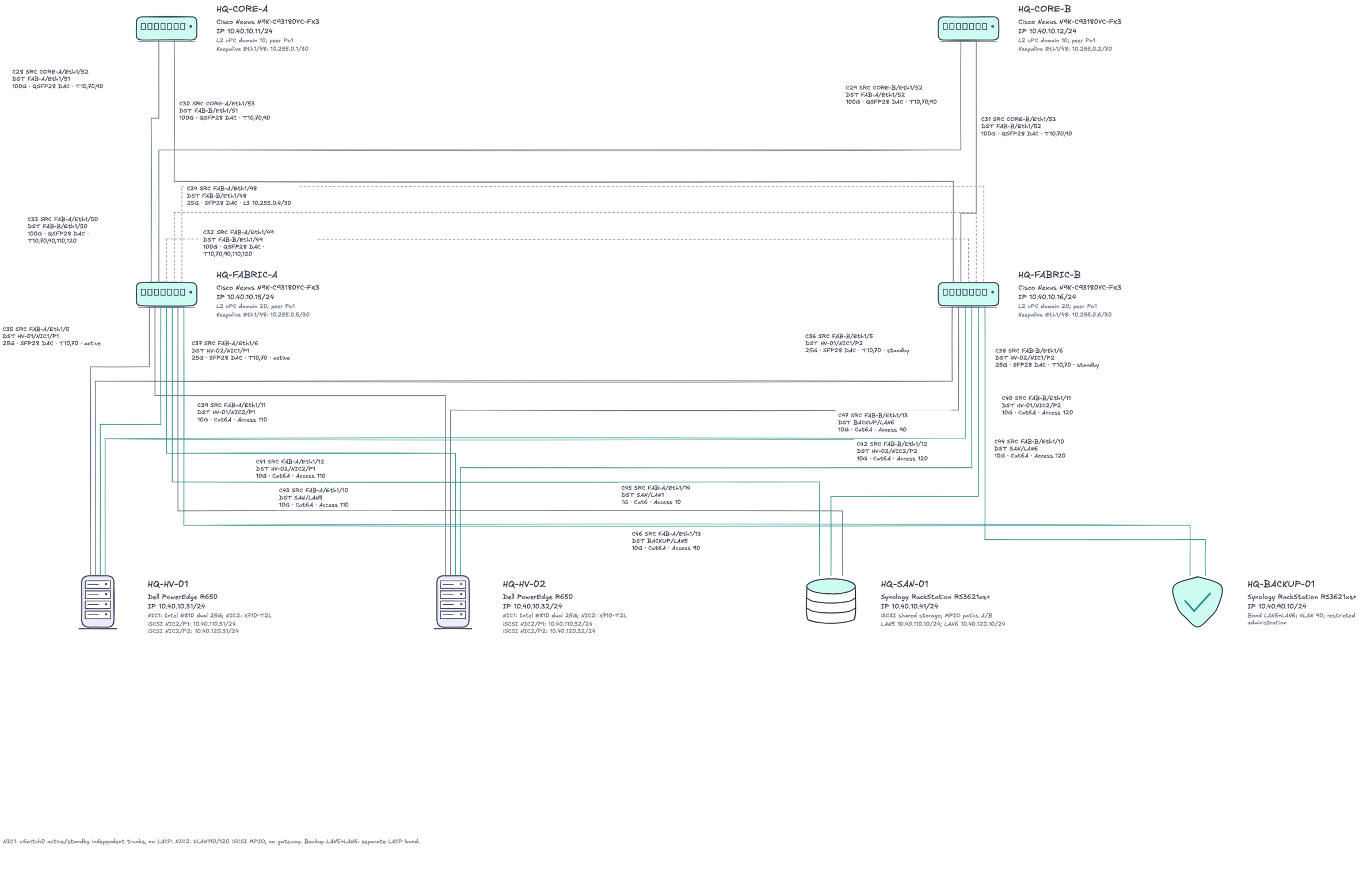
Every displayed endpoint has an exact model and IP · PoE and VLAN intent accompany each physical port label



Fictional reference deployment · every cable has SRC/DST ports · repeat symbols refer to the same inventoried device

AURORA / HQ compute & storage

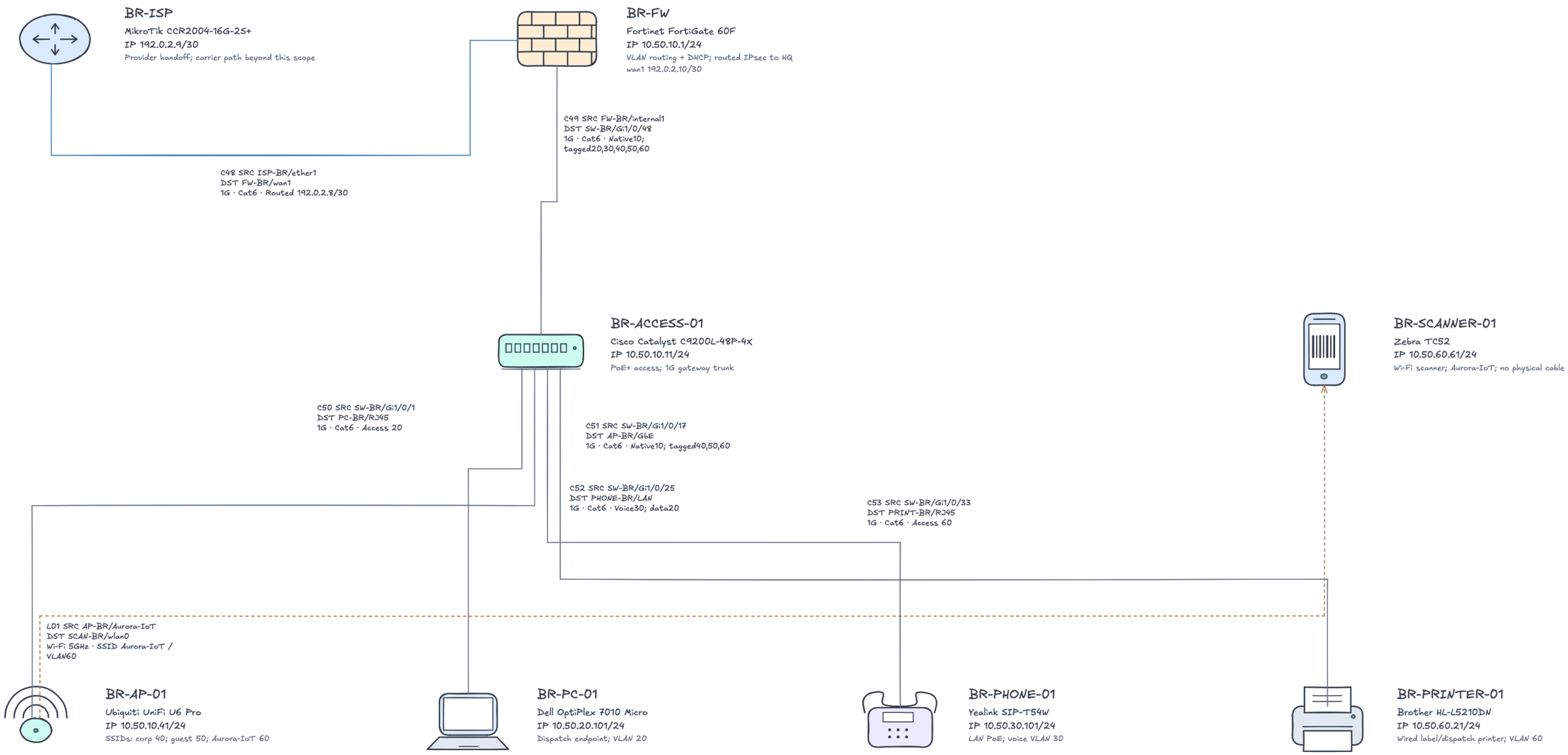
Dual 100G fabric uplinks · 25G host data · two isolated 10G iSCSI paths · exact physical NIC ports



NIC1: vSwitch0 active/standby independent trunks, no LACP. NIC2: VLAN110/120 iSCSI MPIO, no gateway. Backup LAN5+LAN6: separate LACP bond.

AURORA / Québec distribution

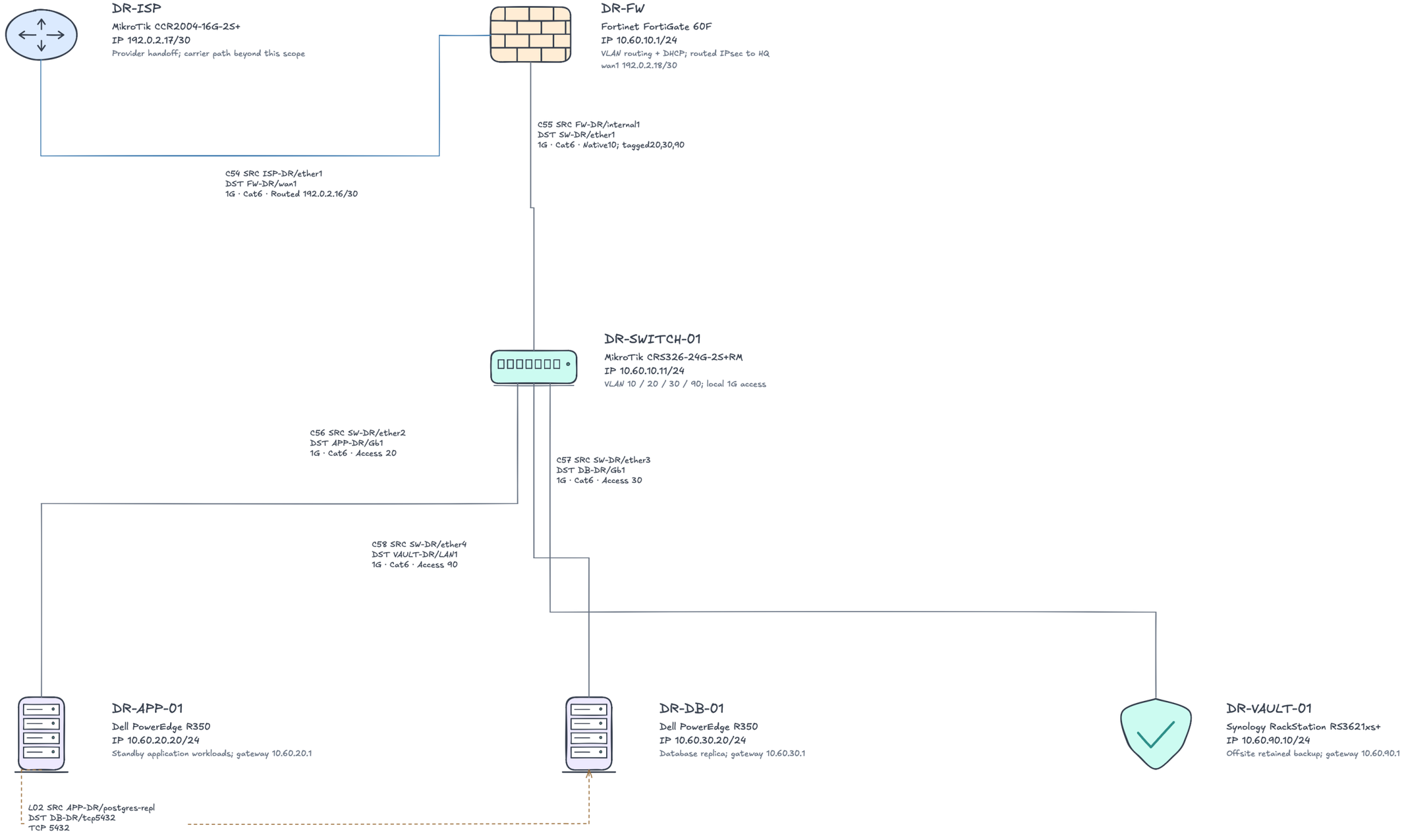
Exact WAN, gateway, switch and endpoint ports · warehouse Wi-Fi is explicitly a logical link



Fictional reference deployment · every cable has SRC/DST ports · repeat symbols refer to the same inventoried device

AURORA / private DR site

Standalone WAN and local VLAN routing · application, database and retained backup have separate addresses



DR has a single gateway/switch and one WAN: documented remaining failure points. Database replication is routed traffic, not a direct Ethernet cable.

AURORA / VPN & replication overlay

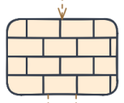
Dashed links are logical interfaces and protocols; C01–C58 on the other sheets are the physical cable plant



REMOTE-01

Dell Latitude 7450
IP 10.40.100.25/24
FortiClient IKEv2; MFA; VPN-assigned address
Outer endpoint 203.0.113.66 (fictional)

L05 SRC REMOTE/ipsec-hq
DST FG-A/ipsec-remote
IPsec IKEv2 / MFA · VPN pool
10.40.100.0/24



HQ-FW-A

Fortinet FortiGate 100F
IP 10.40.10.2/24
Active HA member
Shared WAN VIPs: 198.51.100.10 / 203.0.113.18
Shared VLAN gateways: 10.40.<VLAN>.1

L03 SRC FG-A/ipsec-br
DST FW-BR/ipsec-hq
IPsec IKEv2 · L3 10.254.0.0/30

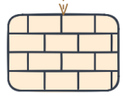
L04 SRC FG-A/ipsec-dr
DST FW-DR/ipsec-hq
IPsec IKEv2 · L3 10.254.0.4/30



HQ-BACKUP-01

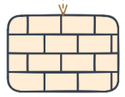
Synology RackStation RS3621xs+
IP 10.40.90.10/24
Bond LAN5+LAN6; VLAN 90; restricted administration

L06 SRC BACKUP/ssh-client
DST VAULT-DR/ssh/tcp22
SSH rsync / TCP22



BR-FW

Fortinet FortiGate 60F
IP 10.50.10.1/24
VLAN routing + DHCP; routed IPsec to HQ
wan1 192.0.2.10/30



DR-FW

Fortinet FortiGate 60F
IP 10.60.10.1/24
VLAN routing + DHCP; routed IPsec to HQ
wan1 192.0.2.18/30



DR-VAULT-01

Synology RackStation RS3621xs+
IP 10.60.90.10/24
Offsite retained backup; gateway 10.60.90.1

HQ HA failover preserves the shared WAN VIPs. Remote outer IP203.0.113.66 is fictitious; the VPN pool is10.40.100.0/24. No diagram line implies a physical cable through the Internet.

AURORA / virtual services

vNIC and port-group names are logical endpoints; the physical hypervisor uplinks are C35–C38 on the compute sheet



HQ-HV-01
Dell PowerEdge R650
IP 10.40.10.31/24
NIC1: Intel E810 dual 25G; NIC2: X710-T2L
iSCSI NIC2/P1: 10.40.110.31/24
iSCSI NIC2/P2: 10.40.120.31/24

L07 SRC HV-01/vSwitch0/PG70
DST DC-01/vNIC0
Virtual switch / VMXNET3 · Port group PG70 / VLAN70

L09 SRC HV-01/vSwitch0/PG70
DST ERP-01/vNIC0
Virtual switch / VMXNET3 · Port group PG70 / VLAN70



HQ-HV-02
Dell PowerEdge R650
IP 10.40.10.32/24
NIC1: Intel E810 dual 25G; NIC2: X710-T2L
iSCSI NIC2/P1: 10.40.110.32/24
iSCSI NIC2/P2: 10.40.120.32/24

L08 SRC HV-02/vSwitch0/PG70
DST DC-02/vNIC0
Virtual switch / VMXNET3 · Port group PG70 / VLAN70

L10 SRC HV-02/vSwitch0/PG70
DST SIEM-01/vNIC0
Virtual switch / VMXNET3 · Port group PG70 / VLAN70



HQ-DC-01
vSphere virtual machine / VMXNET3
IP 10.40.70.10/24
Directory / DNS; preferred resolver
Default gateway 10.40.70.1



HQ-ERP-01
vSphere virtual machine / VMXNET3
IP 10.40.70.20/24
ERP / internal application
Default gateway 10.40.70.1



HQ-DC-02
vSphere virtual machine / VMXNET3
IP 10.40.70.11/24
Directory / DNS; secondary resolver
Default gateway 10.40.70.1



HQ-SIEM-01
vSphere virtual machine / VMXNET3
IP 10.40.70.30/24
Central logs / security monitoring
Default gateway 10.40.70.1

Both directory instances have separate IPs and host placement. Backups and DR recovery validation remain separate from VM placement.

AURORA / VLANs, gateways & deployment notes

All allocations are fictitious · HQ firewall cluster owns the .1 gateway VIPs · branch and DR have local gateways

Site	VLAN	IPv4 subnet	Gateway	Purpose	Policy / operation
HQ	10	10.40.10.0/24	10.40.10.1	Management	Admin jump host + MFA only
HQ	20	10.40.20.0/24	10.40.20.1	Wired clients	DHCP; directory/DNS + business services
HQ	30	10.40.30.0/24	10.40.30.1	Voice	LLDP-MED; PBX / SIP only
HQ	40	10.40.40.0/24	10.40.40.1	Corporate Wi-Fi	802.1X; business services
HQ	50	10.40.50.0/24	10.40.50.1	Guest Wi-Fi	Internet only; deny private destinations
HQ	60	10.40.60.0/24	10.40.60.1	IoT / cameras / printers	DNS/NTP + allowlisted services
BR	10	10.50.10.0/24	10.50.10.1	Management	Admin jump host + MFA only
BR	20	10.50.20.0/24	10.50.20.1	Wired clients	DHCP; directory/DNS + business services
BR	30	10.50.30.0/24	10.50.30.1	Voice	LLDP-MED; PBX / SIP only
BR	40	10.50.40.0/24	10.50.40.1	Corporate Wi-Fi	802.1X; business services
BR	50	10.50.50.0/24	10.50.50.1	Guest Wi-Fi	Internet only; deny private destinations
BR	60	10.50.60.0/24	10.50.60.1	IoT / cameras / printers	DNS/NTP + allowlisted services
HQ	70	10.40.70.0/24	10.40.70.1	Server workloads	Directory .10/.11; ERP .20; SIEM .30
HQ	80	10.40.80.0/24	10.40.80.1	DMZ	HTTPS reverse proxy only
HQ	90	10.40.90.0/24	10.40.90.1	Backup	Restricted admin + offsite transfer
HQ	110	10.40.110.0/24	None (L2)	iSCSI path A	L2 only; MPIO; no routed gateway
HQ	120	10.40.120.0/24	None (L2)	iSCSI path B	L2 only; MPIO; no routed gateway
DR	10	10.60.10.0/24	10.60.10.1	Management	Restricted administration
DR	20	10.60.20.0/24	10.60.20.1	Standby application	To DB TCP5432 only
DR	30	10.60.30.0/24	10.60.30.1	Database replica	Replication allowed from DR app
DR	90	10.60.90.0/24	10.60.90.1	Retained offsite backup	SSH transfer over HQ IPsec; destination retention

WAN handoffs
HQ A 198.51.100.8/30
GW .9 / HA VIP .10
HQ B 203.0.113.16/30
GW .17 / HA VIP .18
Branch 192.0.2.8/30
GW .9 / FW .10
DR 192.0.2.16/30
GW .17 / FW .18

Routed overlays
L03 HQ/BR: 10.254.0.0/30
HQ .1 / branch .2
L04 HQ/DR: 10.254.0.4/30
HQ .5 / DR .6
Remote pool: 10.40.100.0/24
HQ gateway .1; client .25

Deployment checks
1. Validate ports and optics.
2. Stage VLANs and LACP.
3. Test HA + each cable failure.
4. Verify MPIO on both paths.
5. Test guest/IoT isolation.
6. Restore a retained backup.

Known remaining failure points
Branch and DR: one firewall, one switch and one WAN each.
DMZ reverse proxy: one host.
Drawn user endpoints are representative, not a complete fleet BOM.

Port inventory, cable schedule, models and addressing are retained in the accompanying network JSON/CSV; diagrams are directly editable in Excalidraw.